



**Šiaulių
akademija**

VILNIUS UNIVERSITY ŠIAULIAI ACADEMY

BACHELOR PROGRAMME SOFTWARE ENGINEERING

Reports: PHP Final

Object Oriented Programming

Student

Md Sujan Sarkar

Lecturer

Donatas Dervinis

1. Introduction

This project is a PHP object-oriented password manager system. The system allows a user to register, log in, generate secure passwords, and save password records for websites or programs. The saved passwords are stored in a MySQL database in encrypted form.

The project was developed using PHP, MySQL, HTML forms, PDO database connection, password hashing, and AES encryption.

2. Project Purpose

The purpose of this project is to create a PHP OOP program for generating and storing passwords in a database. The program allows users to create an account, log in securely, generate passwords using selected parameters, and save passwords for different services such as Gmail, Facebook, or other websites.

The login password is stored in hashed format, while saved website passwords are encrypted before being stored in the database.

3. Technologies Used

The following technologies were used in this project:

- PHP
- MySQL
- XAMPP
- phpMyAdmin
- HTML
- PDO
- Object-Oriented Programming
- AES-256-CBC encryption
- PHP `password_hash()` and `password_verify()`

4. System Features

The system includes the following features:

- User registration
- User login
- Login password hashing
- User encryption key generation
- Password generation using selected character quantities

- Saving generated or desired passwords
- AES encryption for saved passwords
- Viewing saved passwords after login
- Changing login password
- Re-encryption of the user key after password change

5. Database Structure

The project uses a MySQL database named `password_manager`.

There are two main tables:

users table

The `users` table stores user account information.

Fields:

- `id`: Primary key
- `username`: User login name
- `password_hash`: Hashed login password
- `encrypted_key`: User encryption key encrypted by the login password
- `created_at`: Account creation date and time

password_records table

The `password_records` table stores saved password records.

Fields:

- `id`: Primary key
- `user_id`: Foreign key connected to the `users` table
- `service_name`: Website or program name
- `encrypted_password`: Saved password encrypted with AES
- `created_at`: Password record creation date and time

One user can have many password records.

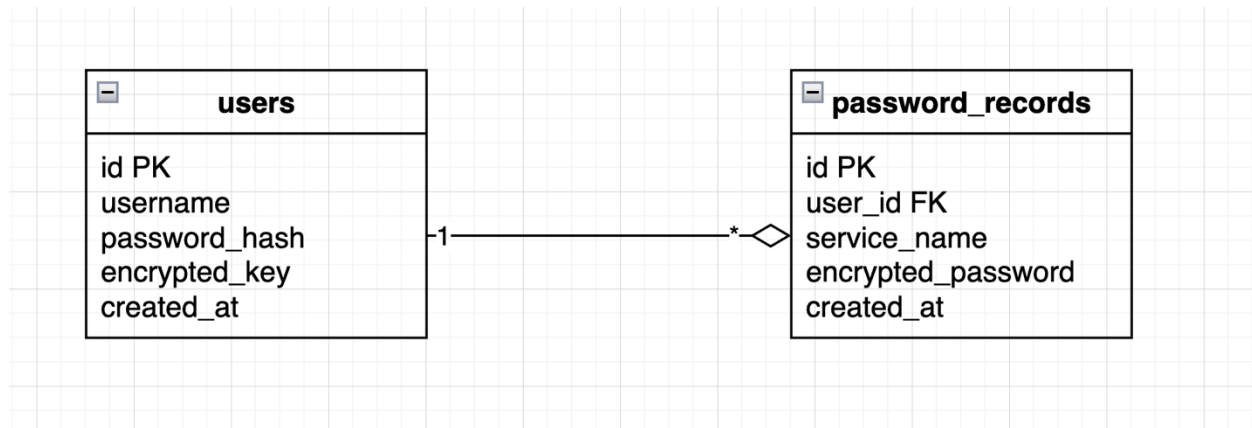


Fig 1: Database

6. UML Class Diagram

The project uses four main classes:

Database class

The `Database` class is responsible for connecting the PHP application to the MySQL database using PDO.

User class

The `User` class is responsible for user registration, login, password hashing, user key encryption, and login password changing.

PasswordGenerator class

The `PasswordGenerator` class is responsible for generating passwords according to selected parameters such as lowercase letters, uppercase letters, numbers, and special characters.

PasswordRecord class

The `PasswordRecord` class is responsible for encrypting saved passwords, saving password records into the database, reading saved records, and decrypting them for the logged-in user.

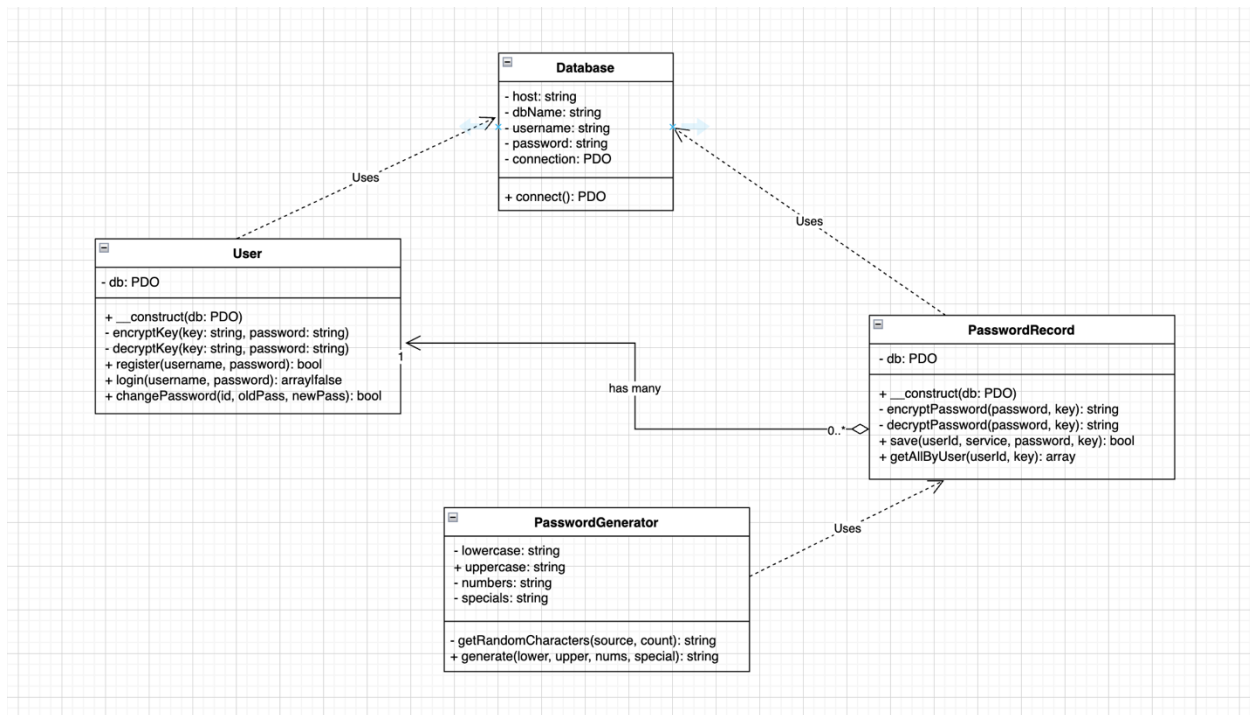


Fig 2: UML Class

7. Application Screenshots

Register Page

This page allows a new user to create an account.

Chrome File Edit View History Bookmarks

localhost/password-manager/ x +

localhost/password-manager/public/reg

Apps Translate VU Timetables | Š... My n

Register

Registration successful. You can now log in.

Username:

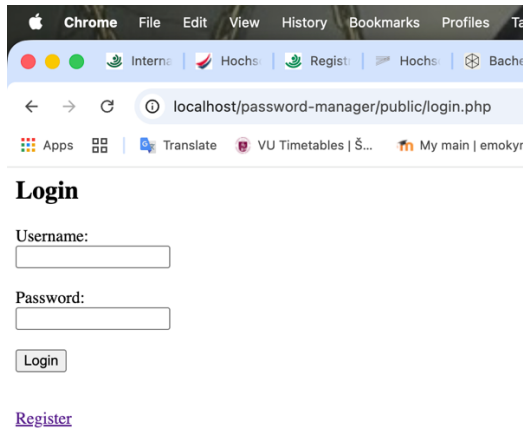
Password:

Register

[Login](#)

Login Page

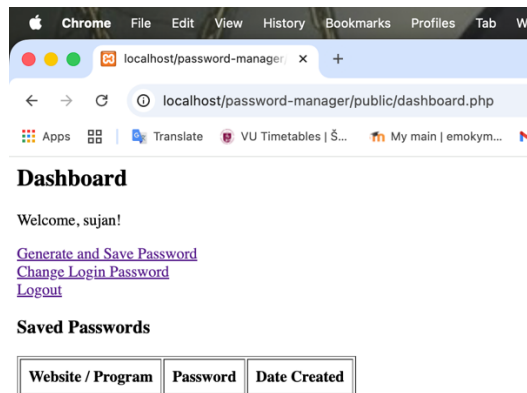
This page allows an existing user to log in using username and password.



The screenshot shows a Chrome browser window with the address bar displaying `localhost/password-manager/public/login.php`. The page has a title "Login". It contains two input fields: "Username:" and "Password:". Below the password field is a "Login" button. At the bottom, there is a link labeled "Register".

Dashboard Page

The dashboard shows the logged-in user and displays saved password records.



The screenshot shows a Chrome browser window with the address bar displaying `localhost/password-manager/public/dashboard.php`. The page has a title "Dashboard". It starts with a welcome message "Welcome, sujan!". Below this are three links: "Generate and Save Password", "Change Login Password", and "Logout". Under the heading "Saved Passwords", there is a table with three columns: "Website / Program", "Password", and "Date Created".

Website / Program	Password	Date Created
-------------------	----------	--------------

Generate Password Page

This page allows the user to generate a password by selecting the number of lowercase letters, uppercase letters, numbers, and special characters.

Generate Password

Lowercase quantity:

Uppercase quantity:

Numbers quantity:

Special characters quantity:

Generated Password:

T8?V0e(Td

Save Password

Website / Program Name:

Password:

[Back to Dashboard](#)

Change Login Password Page

This page allows the user to change the login password. The permanent user key is not changed, but it is re-encrypted using the new login password.

Change Login Password

Password change failed. Old password is incorrect.

Old Password:

New Password:

[Login](#)
[Back to Dashboard](#)

MySQL Users Table

This screenshot shows that the user password is stored as a hash and not as plain text.

Showing rows 0 - 1 (2 total, Query took 0.0001 seconds.)

```
SELECT * FROM `users`
```

Extra options

	id	username	password_hash	encrypted_key	created_at
☐	1	student	\$2y\$10\$tL9e7sJA0FKGB/pj.uzekUPL6N5odrSCAyLr1Deh...	YoeOnVgtocHn1ukK1nTUub+my3l3BgrWir2wHekLsona02c...	2026-05-29 12:11
☐	2	sujan	\$2y\$10\$FUSVsWNCNzVKPoKinvdQJ4z9MxYaH4REYdDmsSHOT...	k8b5tbw5JCLFdiQM5Q/UTGZ/RxYd3l54VkezzOtu6WY5P0j...	2026-05-29 21:41

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

MySQL Password Records Table

This screenshot shows that saved passwords are stored in encrypted format and not as plain text.

Showing rows 0 - 1 (2 total, Query took 0.0001 seconds.)

```
SELECT * FROM `password_records`
```

Extra options

	id	user_id	service_name	encrypted_password	created_at
☐	1	1	Gmail	5H82zRpYzuF34kmSUrDqxHexKnG2PLarIP4mMuPj8s=	2026-05-29 12:34:22
☐	2	2	room	RgNte2GeUM99jT82Bqqvqb7MISvBNeGRKasMMxf4=	2026-05-29 21:47:18

Query results operations

Print Copy to clipboard Export Display chart Create view

8. Code Examples

Database Connection

The database connection is created using PDO in the `Database` class. This makes the database connection reusable in other classes.

User Registration

During registration, the login password is hashed using `password_hash()`. A permanent user key is generated and encrypted using the user login password.

User Login

During login, the password is checked using `password_verify()`. If the password is correct, the encrypted user key is decrypted and stored in the session.

Password Generation

The password generator creates passwords using selected quantities of lowercase letters, uppercase letters, numbers, and special characters.

Password Saving

Before saving a website password to the database, the password is encrypted using AES encryption. The encrypted value is stored in the `password_records` table.

9. Testing

The system was tested with the following steps:

1. A new user account was registered.
2. The user was checked in the `users` table.
3. The login password was confirmed to be stored as a hash.
4. The user logged in successfully.
5. A password was generated using selected parameters.
6. The generated password was saved for Gmail.
7. The saved password appeared on the dashboard.
8. The database was checked and the saved password was encrypted.
9. The login password was changed.
10. The user logged in with the new password.
11. The saved Gmail password was still visible after the login password change.

The testing showed that registration, login, password generation, password saving, encryption, and password changing worked successfully.

10. Conclusion

This project successfully implements a PHP OOP password manager system. It uses object-oriented programming, MySQL database storage, password hashing, AES encryption, and password generation. The system allows users to securely register, log in, generate passwords, save encrypted password records, and change the login password while keeping the same permanent encryption key.

The project meets the main requirements of the task and demonstrates the use of PHP classes, methods, database relationships, encryption, and user interface forms.